

CS677-Spring2025 CRN#22550: MACHINE LEARNING

Assignment#1 - TOTAL POINTS: 100

DUE DATE: 03/13/2025 (Mar 13th)

(This is an Individual assignment)

S.N o.	Questions	Poin ts	Self- Assessm ent																																										
1	Create a graph showing the number of hours studied and the grades earned using the below information (Take hours on X-axis and grades on Y-axis). Describe how studying hours affect grades. <table><tr><th>Hours</th><th>Grades Points</th></tr><tr><td>5</td><td>9</td></tr><tr><td>2</td><td>6</td></tr><tr><td>1</td><td>4</td></tr><tr><td>4</td><td>7</td></tr><tr><td>3</td><td>5.6</td></tr></table>	Hours	Grades Points	5	9	2	6	1	4	4	7	3	5.6	20																															
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2	Read the following Problem Statement [Credit-worthiness assessment]- Predict the probability of a customer being a potential defaulter, based on past transaction data and historical information about customers who have defaulted/not defaulted in their payments. Now answer these questions: i) What type of Machine Learning does this problem belong to? How? ii) Referring to the Week#2 topic Formal Learning stages, list all the steps required to implement the ML solution. (ML algorithm name not required) iii) If your ML solution doesn't perform as expected what do you think might be a reason? Any ideas about how to deal with a model like that?	30																																											
3	Given a Dataset below frame one predictive statement for regression and one for classification. <table><tr><th>Car_Name</th><th>Year</th><th>Selling_Price</th><th>Present_Price</th><th>Kms_Driven</th><th>Fuel_Type</th><th>Sell</th></tr><tr><td>ritz</td><td>2014</td><td>3.35</td><td>5.59</td><td>27000</td><td>Petrol</td><td>Dea</td></tr><tr><td>sx4</td><td>2013</td><td>4.75</td><td>9.54</td><td>43000</td><td>Diesel</td><td>Dea</td></tr><tr><td>ciaz</td><td>2017</td><td>7.25</td><td>9.85</td><td>6900</td><td>Petrol</td><td>Dea</td></tr><tr><td>wagon r</td><td>2011</td><td>2.85</td><td>4.15</td><td>5200</td><td>Petrol</td><td>Dea</td></tr><tr><td>swift</td><td>2014</td><td>4.60</td><td>6.87</td><td>42450</td><td>Diesel</td><td>Dea</td></tr></table>	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Sell	ritz	2014	3.35	5.59	27000	Petrol	Dea	sx4	2013	4.75	9.54	43000	Diesel	Dea	ciaz	2017	7.25	9.85	6900	Petrol	Dea	wagon r	2011	2.85	4.15	5200	Petrol	Dea	swift	2014	4.60	6.87	42450	Diesel	Dea	20	
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4	What are the names of the two loss functions L_S and L_D that we discussed in week#2 class time? Explain in detail with examples. What is the topic connecting these two loss functions and give the equation for the same? (Your answer must be at least 50 words)	30	
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Submission:

- Submit a Word document with a detailed explanation and draw diagrams wherever necessary. You can also write and draw on paper, and scan to submit (make sure your scanned copies are clear and readable or you will have a points deduction).
- Write your **Name** and **UID** on your submission document.
- You may not be graded full points if your submissions are not clear or unreadable.
- After the **Due Date**, late submission of up to one week will incur a 10% loss of total points. 5% every week thereafter until the end date.
- Attach this file with self-assessment. This is for your reference to check if you answered all the questions completely before submission.
- A graph template will be attached additionally for plotting Question 1.